

# Complete Project Report

## AI Student Helpdesk Assistant

Academic Project Report

Submitted in partial fulfillment of the requirements for the degree of  
[Degree Name] in [Department Name]

Submitted By:

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Under the Guidance of:

[Guide Name]

[Guide Designation]

[College Name]

[Academic Year]

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### DECLARATION

We hereby declare that the project entitled "AI Student Helpdesk Assistant" submitted to [College Name] is a record of an original work done by us under the guidance of [Guide Name].

### ACKNOWLEDGEMENT

We express our sincere gratitude to [Guide Name] for their invaluable guidance.

### ABSTRACT

The AI Student Helpdesk Assistant is a centralized academic management and self-service portal designed to streamline information retrieval in educational institutions. The system replaces manual enquiry processes with an intelligent Extractive QA Chatbot (built using TF-IDF and Cosine Similarity) that provides instant answers from college-uploaded documents (PDF/DOCX), alongside real-time updates on assignments, timetables, and announcements.

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## **Chapter 1: Introduction**

### **1.1 Background**

Educational institutions generate vast amounts of operational and academic information. Students often struggle to find answers to routine questions, leading to administrative overhead.

### **1.2 Problem Statement**

Information is distributed across notice boards, disparate documents, and departments. Finding academic information is time-consuming, resulting in repeated queries to the administration.

### **1.3 Project Objectives**

- Centralize academic information (timetables, assignments, announcements).
- Provide a Student Self-Service Portal.
- Implement an AI-based chatbot using NLP techniques (TF-IDF) to extract answers from uploaded PDFs and DOCX files.
- Deliver an Administrative Dashboard for real-time analytics.

### **1.4 Project Scope**

The project covers Admin capabilities (managing students, subjects, materials, etc.) and Student capabilities (interacting with the AI, viewing timetables/assignments). It operates offline using a local database and ML libraries, strictly adhering to zero-internet-dependency rules.

### **1.5 Limitations**

- The AI relies on Extractive QA (TF-IDF), so it extracts verbatim text rather than generating new contextual sentences (LLM).
- Scalability is bounded by SQLite capabilities.

## **Chapter 2: Existing and Proposed System**

### **2.1 Existing System**

- Manual student enquiry handling.
- Information distributed across notices, documents, and departments.
- Repeated questions to college administration.

## 2.2 Proposed System

- Centralized Academic Information: A single digital repository for college data.
- Student Self-Service Portal: View profiles, timetables, and assignments.
- AI-based Question Answering: An offline chatbot that parses college-uploaded study materials to answer queries instantly.
- Administrative Information Management: A Next.js dashboard for admins (including Faculty and Announcement management).
- Dynamic Theming Engine: A robust Light/Dark mode and accent color system customizable per user.